A vertical plane passing through magnetic axis is called magnetic meridian and a great circle perpendicular to Earth's magnetic axis is called magnetic equator.

When a magnetic needle is freely suspended, the alignment of the magnet does not exactly lie along the geographic meridian as shown in Figure 3.4. **The angle between magnetic meridian at a point and geographical meridian is called the *declination or magnetic declination*** (D). At higher latitudes, the declination is greater whereas near the equator, the declination is smaller. In India, declination angle is very small and for Chennai, magnetic declination angle is $-1^{\circ}16'$ (which is negative (west)).

The angle subtended by the Earth's total magnetic field $\vec{B}$ with the horizontal direction in the magnetic meridian is called dip or magnetic inclination (I) **at that point** (Figure 3.5). For Chennai, inclination angle is $14^{\circ}28'$. **The component of Earth's magnetic field along the horizontal direction in the magnetic meridian is called horizontal component of Earth's magnetic field, denoted by B_H.**

Let B_E be the net Earth's magnetic field at any point on the surface of the Earth. B_E can be resolved into two perpendicular components.

$$\text{Horizontal component} \quad B_H = B_E \cos I \qquad (3.1)$$

$$\text{Vertical component} \quad B_V = B_E \sin I \qquad (3.2)$$

Dividing equation (3.2) and (3.1), we get

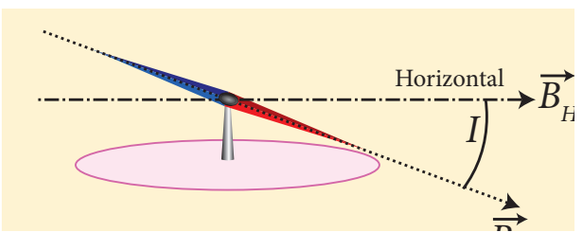


Figure 3.5 Inclination angle

$$\tan I = \frac{B_V}{B_H} \qquad (3.3)$$

(i) At magnetic equator

The Earth's magnetic field is parallel to the surface of the Earth (i.e., horizontal) which implies that the needle of magnetic compass rests horizontally at an angle of dip, $I = 0^{\circ}$.

$$B_H = B_E$$
$$B_V = 0$$

This implies that the horizontal component is maximum and vertical component is zero at the equator.

(ii) At magnetic poles

The Earth's magnetic field is perpendicular to the surface of the Earth (i.e., vertical) which implies that the needle of magnetic compass rests vertically at an angle of dip, $I = 90^{\circ}$. Hence,

$$B_H = 0$$
$$B_V = B_E$$

This implies that the vertical component is maximum at poles and horizontal component is zero at poles.

EXAMPLE 3.1

The horizontal component and vertical component of Earth's magnetic field at a place are 0.15 G and 0.26 G respectively. Calculate the angle of dip and resultant magnetic field. (G-gauss, cgs unit for magnetic field $1\text{G} = 10^{-4}$ T)

Solution:

$B_H = 0.15$ G and $B_V = 0.26$ G

$$\tan I = \frac{0.26}{0.15} \Rightarrow I = \tan^{-1}(1.732) = 60^{\circ}$$

The resultant magnetic field of the Earth is

$$B = \sqrt{B_H^2 + B_V^2} = 0.3 \text{ G}$$

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